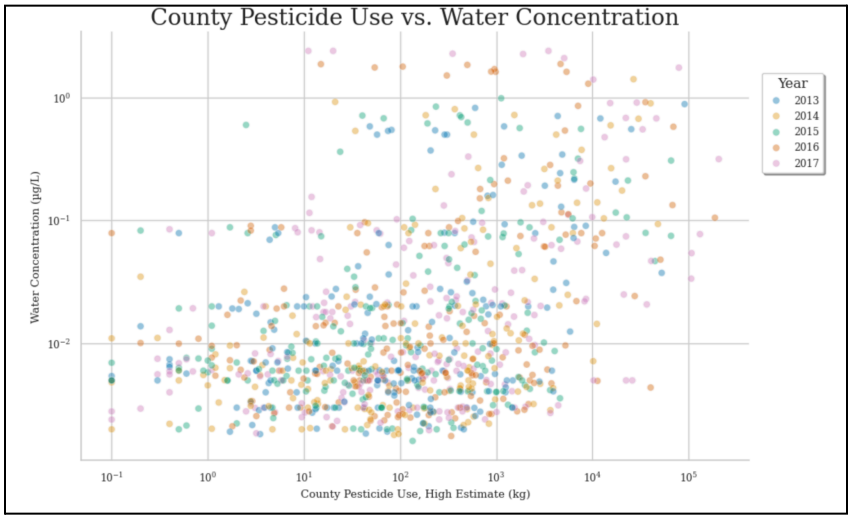
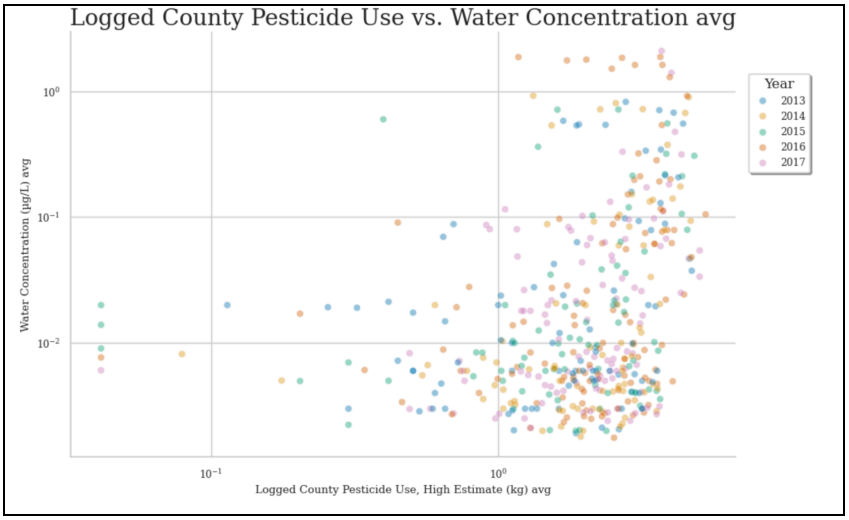
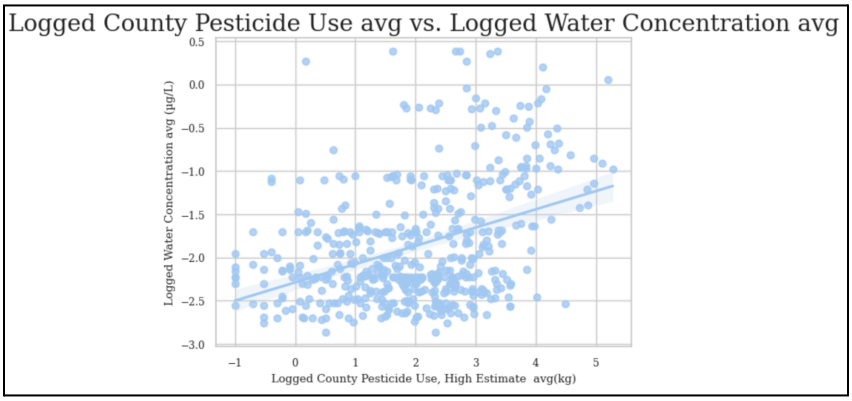
Question: Do regions with higher pesticide use have higher contamination concentrations in surface water?

To address this question, we plotted estimated pesticide use (kg) by measured pesticide concentrations (µg/L) in surface waters. We initially plotted the raw values, then reproduced the plot with the estimated pesticide use on a log scale, and finally reproduced the plot with both the pesticide usage and contaminant concentrations on a log scale. This final transformation revealed a more apparent trend in the data, suggesting that pesticide contaminant concentrations increase with pesticide usage. A limitation of this figure is that it does not give any information about the pesticide categories.

To begin our initial analytical approach, we created a linear regression model. We felt that this approach was best suited for our data, as we are investigating the relationship between two variables. The visual representation of a linear regression clearly asserts whether or not a trend exists within the data, and can be supported with additional statistics. In this case, we are using the R^2 and p-value to determine the strength of the hypothesized relationship. A p-value of <0.001 suggests that there is some correlation between pesticide usage and contamination. An R^2 value of 0.14 indicates that county pesticide usage explains 14% of the variation in pesticide concentrations in surface waters. While this suggests that a majority of the variation in surface waters is driven by other factors, such as rainfall, watershed, topography, soil characteristics, etc., it does account for a portion of the contamination variation.



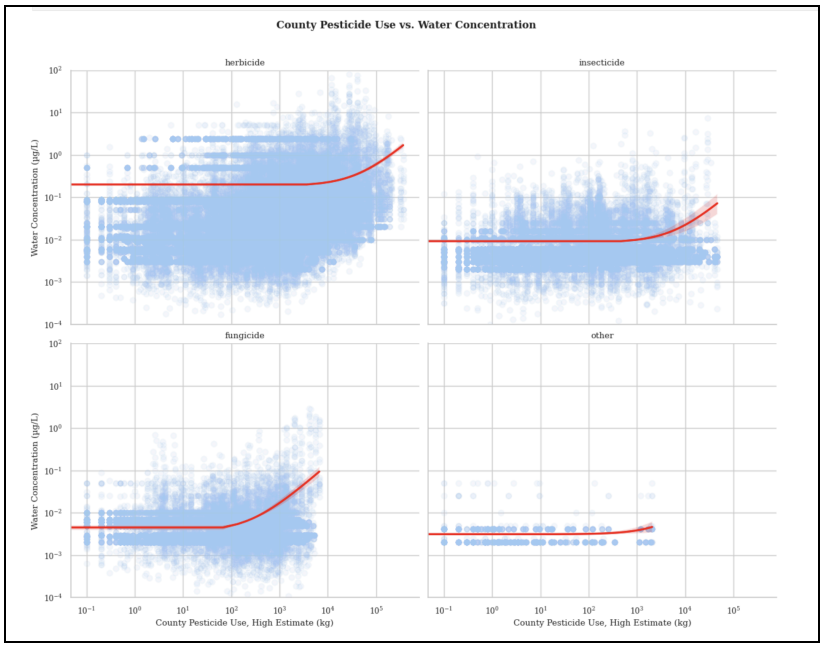


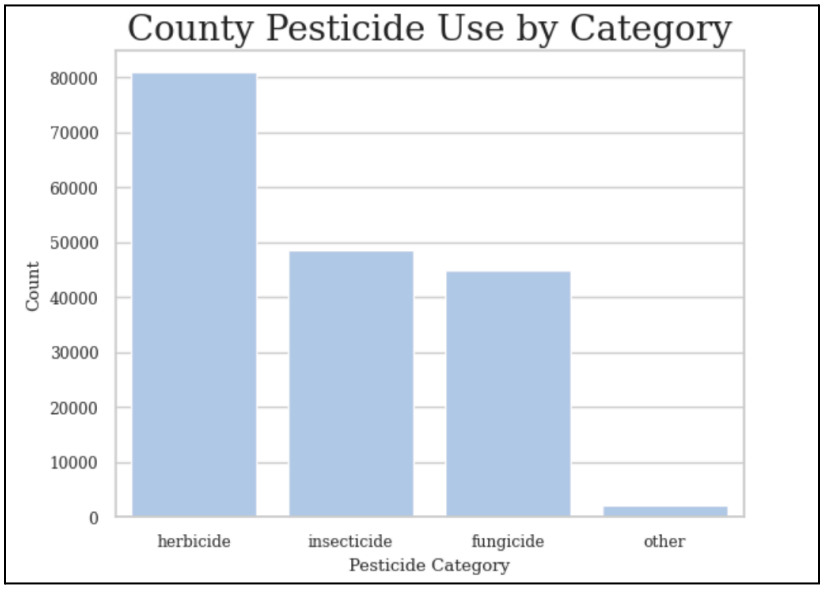



```
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intercept     -2.28161
R²             0.13747
p-value        0.00000
dtype: float64
```

Question: Are there stronger correlations of pesticide use and contamination among different pesticide categories (herbicides, fungicides, insecticides, etc.)?

To address this question, we first investigated the frequency of use of each pesticide category, then compared the usage vs. contaminant concentration. The x-axis of the bar graph lists the pesticide category while the y-axis is a count of how frequently that pesticide occurred in our data set. While this visualization suggests that herbicides are used more frequently than the other pesticides, a limitation of this graph is the fact that it does not reveal information about the mass of pesticide used, which could be misleading. The categorized plot compares the relationships between estimated usage (kg) and water contamination (µg/L) of each pesticide category. This visualization is not very informative and is limited by the sheer amount of data present. To further investigate this, we plan to average the data by county (rather than individual sample collection sites) and group by each year to reduce the volume of data being displayed.





How has pesticide use and cumulative pesticide contamination changed over time?
To begin investigating this question, we plotted the frequency of occurrences within each pesticide category in our dataset from 2013 to 2017. To further explore the potential relationship between pesticide use and contamination, we plan to plot the total estimated mass of pesticides used in each category per year, along with the average contaminant concentration for each category per year. While the previous figures examine whether a relationship exists between pesticide use and contamination, analyzing how these variables change over time may provide additional evidence of a potential association. However, an important limitation of this

investigation is the previously calculated R^2 value, which indicates that pesticide usage explains only 14% of the variation in contaminant concentrations. Although the proposed plots may reveal apparent trends or associations, they may visually suggest a stronger relationship than the data actually supports.

